

CS 160 — Team 4

Sprint 1 Software Engineering Report

Voice Assistant for Older Adults

San Jose State University, CS 160 Software Engineering, Fall 2026

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Sprint 1: Week 6 – 7

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1. Git Account Profile Member List

Team Member	GitHub Username	Draft Responsibility
Moebius Yang	momoyang0828	Frontend
Zohreh Ashtarilarki	ZohrehAshtarilarki	Backend
Amarjargal Ayurzana	Akiko0210	Backend
Kunj Shah	kunjcr2	AI and voice exploration

2. Survey Report and Key Feature Decision

The survey collected 13 responses. All 13 respondents identified themselves as people who help care for or support someone age 60 or older. Therefore, the evidence is useful for caregiver/supporter priorities but does not directly represent older adults answering for themselves.

The team reviewed both the average usefulness ratings and the Top 3 selections from the survey when selecting the seven features for implementation. The final selection also considered the goals and scope of the project. The voice conversational interface and confirmation-before-action behavior are supporting system requirements used across the selected features and are not counted as separate core features.

Selected feature	Extremely Useful	Top-3 Votes
Doctor Appointment Booking	10	12
Appointment Reminders	8	9
Medication Reminders	9	8
Prescription Refill Assistance	8	0
Human Assistance / Live Agent Handoff	7	1
Multilingual Voice Support	6	3
Messaging a Doctor or Care Team	5	3

2.1. Survey Results

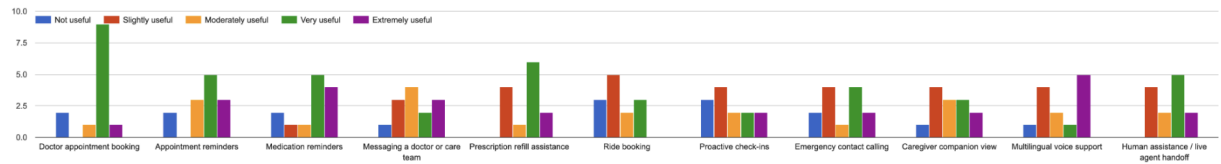


Figure 1. Survey feature usefulness ratings.

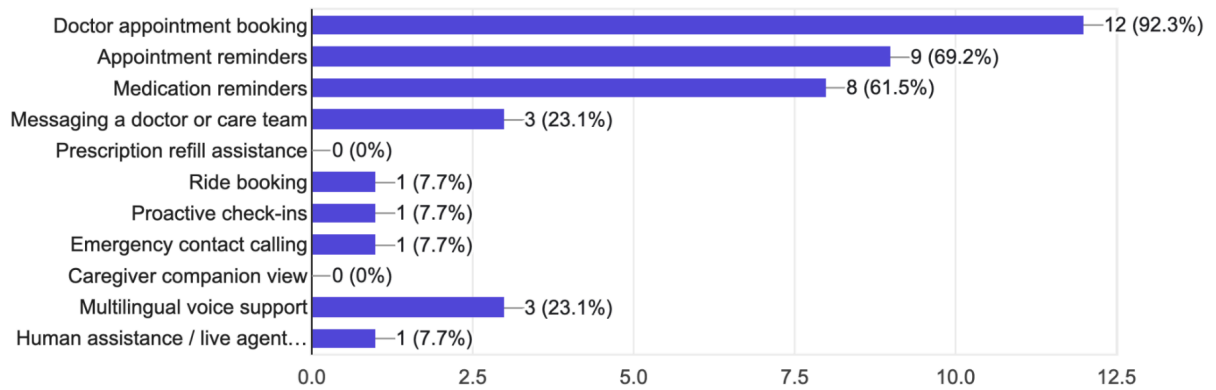


Figure 2. Features selected by respondents as their three most important choices.

Key interpretation: the survey results show that appointment booking, appointment reminders, and medication reminders were the most useful features to respondents. Multilingual support was also important. Eleven out of 13 respondents rated support for their preferred language as important or extremely important, and all 13 said this feature would make them more likely, or possibly more likely, to use the product. Human assistance was also kept because some users wanted to connect directly with a person, while others preferred having a caregiver contacted if the system kept having trouble understanding them.

3. Project Scope

Sprint 1 is a planning and specification sprint. The goal is to finalize the product direction, document the system design, organize the shared repository, and prepare the team to begin implementation.

Sprint 1 Deliverables

1. Finalize the seven core features using the survey results.
2. Define the project scope and supporting system requirements.
3. Complete the architectural block, component, sequence, and class diagrams.
4. Select React, Python with FastAPI, PostgreSQL, OpenAI Whisper for speech-to-text, OpenAI TTS for speech output, and the OpenAI API for AI features.
5. Create and maintain the product backlog, Sprint 1 backlog, task assignments, and sprint log.
6. Organize the shared GitHub repository and Sprint 1 documentation.
7. Record the sprint outcome and prepare the Sprint 2 implementation plan.

Outside the Sprint 1 Scope

- Production healthcare, FHIR, rideshare, payment, pharmacy, or calendar integrations
- Medical advice or emergency detection
- A production-ready voice pipeline or full multilingual support
- Caregiver monitoring features
- Deployment, HIPAA certification, or clinical validation

Selected Technology Stack

The system uses a React frontend, a Python backend with FastAPI, PostgreSQL for persistence, OpenAI Whisper for speech-to-text, OpenAI TTS for speech output, and the OpenAI API for AI-assisted request processing. External healthcare services are represented by mock adapters during early development.

4. High-Level Architectural Block Diagram

MVC mapping: React serves as the View, and FastAPI routes act as the Controllers. The service layer contains the main business logic, while the domain models and repository layer manage application data and communication with PostgreSQL. The ai/ package handles communication with the external AI API and is part of the backend, not a separate server.

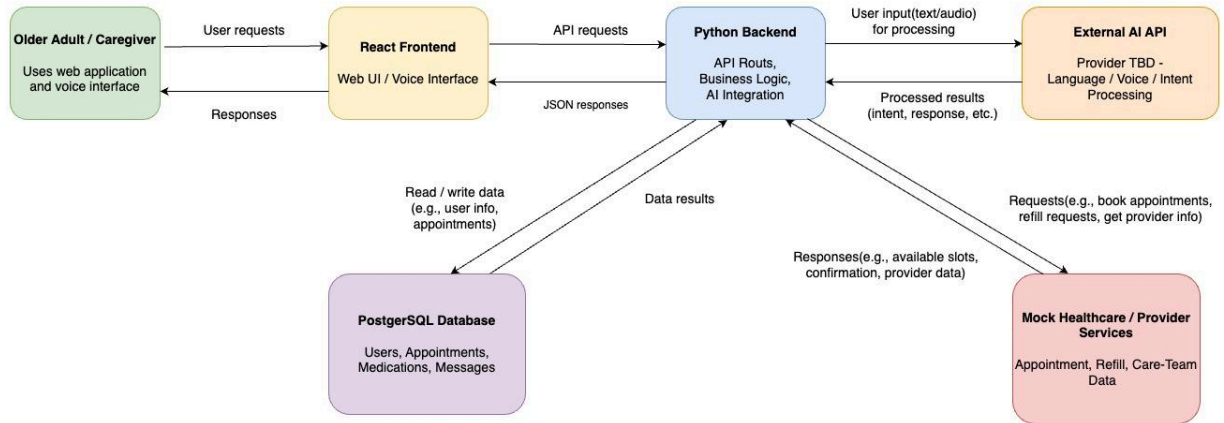


Figure 3. High-level system architecture and integration boundaries.

5. Component Diagram

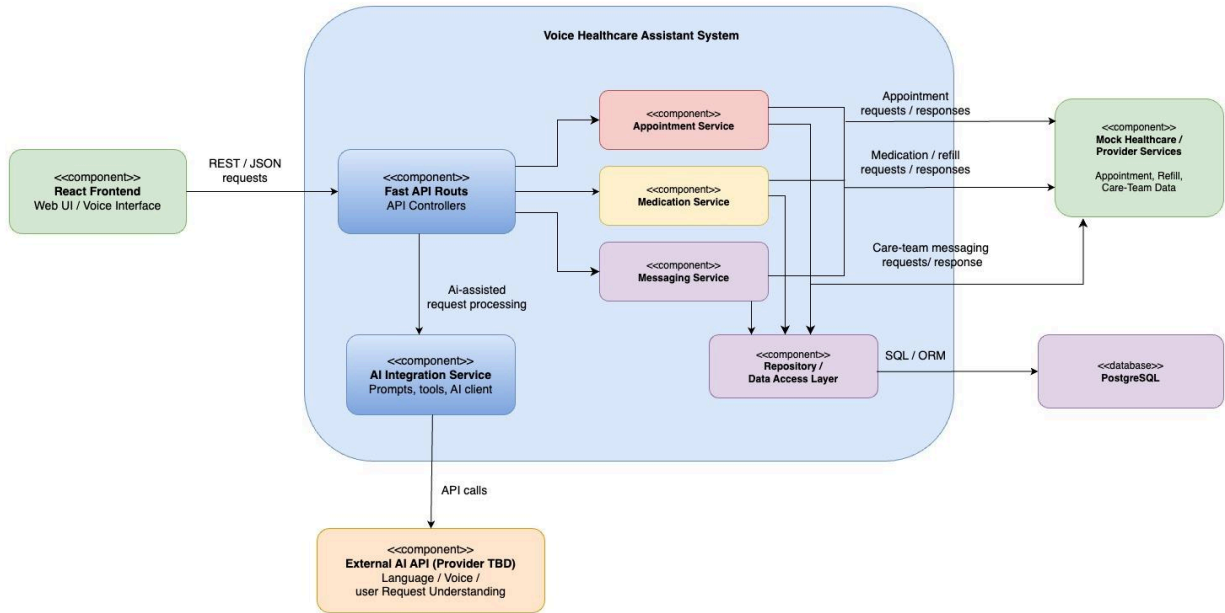


Figure 4. System components for the React frontend, FastAPI backend, AI service, data layer, and provider adapter.

6. Sprint 1 Sequence Diagram

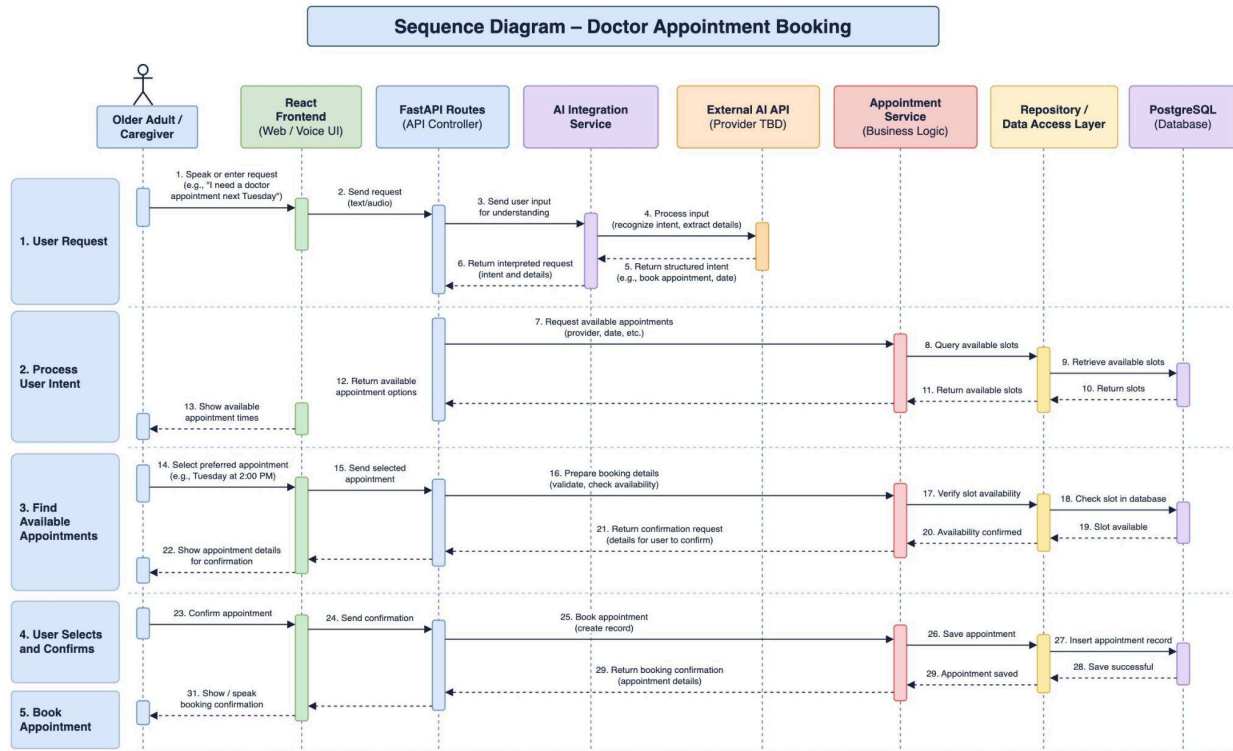


Figure 5. Doctor appointment booking flow from request through explicit confirmation.

7. Class Diagram

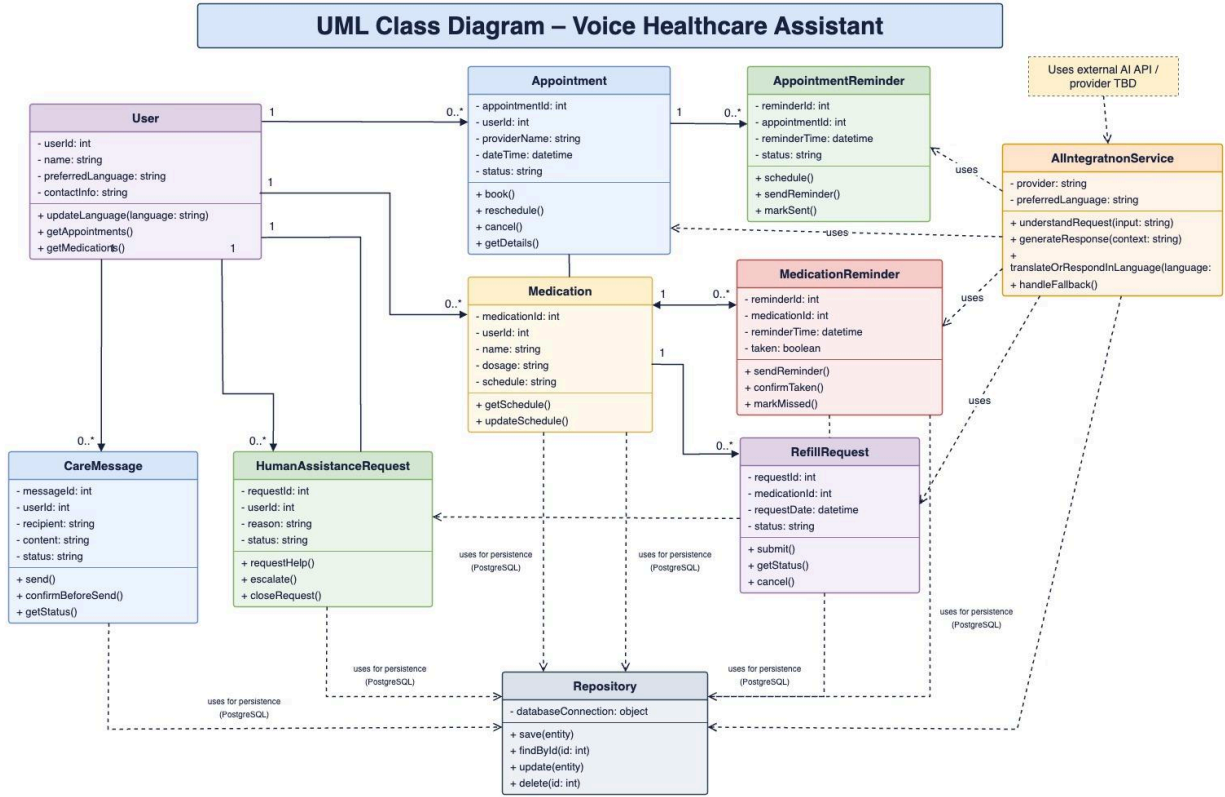


Figure 6. Main domain entities, services, and persistence relationships.

8. Git Source Code

Repository: github.com/kunjcr2/CS160_Project

Main branch: github.com/kunjcr2/CS160_Project/tree/main

Commit history: github.com/kunjcr2/CS160_Project/commits/main

9. Product Backlog and Sprint Log

The Product Backlog is provided as a separate spreadsheet. It contains the main project tasks, including planning, design, setup, and future implementation tasks for all seven selected features. Sprint 1 focuses only on project planning and specification, including finalizing the project scope, creating the required diagrams, preparing the backlogs, assigning team responsibilities, organizing the documentation, and setting up the shared GitHub repository. Feature implementation will begin in later sprints. The Product Backlog will be updated throughout the project as tasks are completed, changed, or moved between sprints. A tentative feature schedule for later sprints will be decided by the team based on priority, workload, and progress after Sprint 1.

[Team 4 Backlog and Sprint Tracking Spreadsheet](#)

10. Sprint Summary

Sprint 1 completed the planning and specification work needed to begin development. The team reviewed 13 caregiver/supporter survey responses, selected seven core product features, defined the initial scope, completed the architecture and design diagrams, created the product and sprint backlogs, assigned work across the four team members, selected the initial technology stack, and organized the shared GitHub repository.

Application implementation was not completed during Sprint 1 and will carry into Sprint 2. The React frontend, FastAPI backend, PostgreSQL schema, mocked appointment workflow, automated tests, and voice pipeline still need to be built. The survey also did not include direct responses from older adults, so the feature decisions remain hypotheses that should be validated with intended users.

What went well was the use of survey evidence to narrow the product scope and the creation of a shared architecture before implementation. The main blockers were limited time, missing direct older-adult feedback, and the need to finalize the technology stack. In Sprint 2, the team should record work in the shared sprint log as it happens, use feature branches and pull requests, and focus on one small end-to-end appointment flow before expanding to other features.

Task assignments, hours, completion dates, statuses, and blockers are maintained in the shared Google Sheet and should be confirmed by each team member before submission.

11. Plans for Sprint 2

The provisional Sprint 2 goal is to turn the mocked appointment flow into a stable vertical slice and begin reminders.

1. Create the React conversation interface and connect it to the Python/FastAPI API.
2. Create the PostgreSQL schema and persist users, appointments, and confirmations.
3. Implement appointment lookup, proposal, confirmation, and cancellation against a mock provider service.
4. Add appointment reminders with configurable timing.
5. Add automated tests for confirmation and error paths.
6. Prototype voice input with OpenAI Whisper and speech output with OpenAI TTS.
7. Conduct usability feedback with at least one intended user if possible.

Open Decisions

- PostgreSQL driver, ORM, and migration tool
- OpenAI model choices and request limits for the first vertical slice
- Authentication approach
- Frontend component and styling approach
- Hosting and deployment platform
- Final actual hours, completion dates, and blockers after team review in the shared Google Sheet